

HIGH PRAIRIE AGDM, AB, Canada

WMO: 712260

Lat: 55.3949N

Lon: 116.4831W

Elev: 591

StdP: 94.42

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-33.6	-29.9	-37.4	0.1	-33.7	-33.4	0.2	-29.8	10.6	4.8	9.3	2.5	1.2	260	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	27.3	16.9	25.6	16.0	24.0	15.0	18.7	24.8	17.4	23.2	16.3	22.0	3.5	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	12.5	21.6	15.1	11.5	20.1	13.9	10.7	18.6	55.4	24.9	51.2	23.2	47.8	21.9	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.5	6.5	DB	-37.7	30.7	4.0	1.6	-40.6	31.8	-43.0	32.7	-45.3	33.6	-48.2	34.8	(4)
(5)			WB	-38.1	20.8	4.0	1.7	-41.0	22.0	-43.3	23.0	-45.6	23.9	-48.5	25.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-11.3	-10.4	-5.1	3.8	10.0	14.3	16.5	15.0	10.6	4.0	-5.0	-12.4
	DBStd	12.15	9.48	8.48	8.19	4.56	4.31	2.85	2.52	3.01	4.08	4.82	7.70	8.52
	HDD10.0	3321	659	571	467	191	52	3	0	2	42	190	452	693
	HDD18.3	5769	917	804	726	437	258	123	67	109	232	443	701	952
	CDD10.0	613	0	0	0	4	53	132	203	156	59	5	1	0
	CDD18.3	19	0	0	0	0	1	3	11	5	0	0	0	0
	CDH23.3	479	0	0	0	4	34	80	204	116	41	1	0	0
	CDH26.7	72	0	0	0	1	4	7	36	18	7	0	0	0
Wind	WSAvg	2.9	2.8	2.6	3.0	3.4	3.4	3.0	2.8	2.6	3.0	3.1	2.8	2.7
Precipitation	PrecAvg	486	27	18	21	24	46	88	85	57	41	28	27	24
	PrecMax	651	59	31	44	45	90	158	147	107	107	65	57	58
	PrecMin	317	7	6	5	4	15	30	47	19	4	13	11	7
	PrecStd	78	13	7	8	9	19	34	28	21	18	12	12	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.8	8.0	12.0	22.6	26.8	27.7	30.0	28.8	27.6	21.4	13.9	8.0
		MCWB	4.0	3.5	6.1	10.1	13.5	15.8	18.7	18.2	16.1	12.3	6.9	4.0
	2%	DB	6.4	5.9	9.4	17.6	23.7	25.7	27.4	26.3	23.9	16.4	9.5	5.3
		MCWB	2.9	2.1	4.2	7.9	12.1	15.0	17.9	17.3	14.4	9.5	4.9	2.0
	5%	DB	4.9	4.0	7.2	14.5	21.5	23.8	25.6	24.3	21.4	13.5	6.5	3.2
		MCWB	2.0	1.0	3.0	6.6	10.8	14.1	16.9	15.9	13.5	7.6	3.2	0.6
	10%	DB	2.7	1.7	5.2	11.9	19.3	22.0	23.9	22.5	18.5	11.5	4.4	0.5
		MCWB	0.4	-0.5	1.7	5.2	9.9	13.3	16.0	15.2	11.7	6.5	1.5	-1.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.4	3.7	6.5	10.7	14.5	17.7	21.0	20.2	17.1	13.0	7.2	4.1
		MCDB	8.2	7.2	11.0	20.6	23.9	24.2	27.0	26.2	25.0	20.5	12.4	7.3
	2%	WB	3.1	2.4	4.6	8.7	12.9	16.2	19.1	18.5	15.5	10.3	5.1	2.3
		MCDB	6.0	5.4	8.7	16.9	21.2	23.0	25.2	24.4	22.7	15.7	8.6	5.0
	5%	WB	2.1	1.2	3.2	6.9	11.7	15.1	17.9	17.1	13.9	8.4	3.4	0.5
		MCDB	4.7	3.9	7.1	13.5	19.5	22.0	23.9	22.5	20.3	12.2	6.2	3.1
	10%	WB	0.5	-0.5	1.8	5.5	10.6	14.2	16.7	15.9	12.5	7.0	1.9	-1.5
		MCDB	2.7	1.7	5.1	11.2	17.9	20.2	22.2	21.1	17.4	10.7	4.1	0.4
Mean Daily Temperature Range	5% DB	MDBR	10.5	11.4	12.0	11.8	13.6	12.3	12.2	12.4	12.5	10.4	9.5	10.2
		MCDBR	10.5	13.0	13.2	17.3	18.0	16.7	16.1	16.5	17.7	14.7	10.9	12.4
		MCWBR	7.9	9.8	8.8	9.3	8.3	7.8	8.0	8.6	9.2	8.8	7.4	10.0
		MCDBR	10.1	12.5	12.8	16.4	16.1	14.1	14.9	15.3	17.0	14.1	10.5	12.5
	5% WB	MCWBR	7.8	9.5	8.5	9.1	7.7	6.9	8.1	8.8	9.2	9.0	7.5	10.4
Clear-Sky Solar Irradiance	taub	0.240	0.252	0.286	0.332	0.347	0.352	0.354	0.350	0.313	0.285	0.252	0.232	
	taud	2.350	2.405	2.406	2.389	2.400	2.406	2.433	2.419	2.535	2.576	2.448	2.283	
	Ebn at Noon	727	857	897	893	895	892	883	864	859	811	713	640	
	Edh at Noon	49	68	87	103	108	109	104	99	77	58	45	41	
All-Sky Solar Radiation	RadAvg	0.78	1.78	3.32	4.70	5.57	6.01	5.90	4.83	3.28	1.78	0.82	0.55	
	RadStd	0.07	0.14	0.27	0.30	0.41	0.46	0.39	0.39	0.28	0.21	0.06	0.05	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+58	N/A

Nomenclature: See separate page